

Doctoral Training Programmes

The titles of the Doctoral School's training programs: (valid from June 26, 2023).

Doctoral School of Informatics/Doctoral Program of Software and Computer Science

Doctoral School of Informatics/Doctoral Program of Data Science, Networks, Information Systems

Doctoral School of Informatics/Doctoral Program of Scientific Computing and Models, Numerical and Symbolic Methods

Doctoral School of Informatics/Doctoral Program of Informatics Teaching Methodology

Doctoral School of Informatics/Doctoral Program of Geospatial Information and Spatial Data

Doctoral School of Informatics/Doctoral Program of IT Solutions in Engineering Sciences

Doctoral Program of Software and Computer Science

Head of the programme: Dr. Zoltán Horváth, professor

The aim of the doctoral program is to provide doctoral students with deep and wide-ranging knowledge that will enable them to cultivate the theoretical foundations of computing and software science in an effective and active way, furthermore, to apply the methodological principles in an innovative manner, and adopt new procedures.

The program offers the following research topics:

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Information Systems

Head of the programme: Dr. András Benczúr, professor emeritus

The doctoral program is concerned with the theoretical research, construction, operation, and use of computer technologies for the operation of the digital data world. The three pillars of the doctoral program, data science, computer networks, and original information systems, support each other and provide a wide range of research opportunities.

The increasingly abundant availability of data and processing capabilities has led to the emergence of the field of data science, enhancing previous data mining and big data technologies with new deep learning capabilities in Artificial Intelligence. In summary, the field of managing, analyzing, and using big data sets is called data science. This includes data collection, cleaning, storage, and related algorithms, as well as data mining and deep learning technologies for analysis.

In addition to data management, the research of digital data networks for data collection, flow, access, and communication is the second component of the program. Networks will connect cloud centers to server centers, end users to individual mobile devices, and sensor networks to the Internet of communicating smart objects, extending the connectivity of our world across a very broad spectrum.

In the area of information systems, the program focuses on new directions in classical system models and operational database management technologies. Research in new concepts for data models, knowledge representation, and new database management technologies is also linked to this area. A key feature of digitization is that information systems are being built and enhanced with increasing automation and integration of AI technologies. Methods, algorithms, and technologies collectively referred to as artificial intelligence, and advanced data analytics solutions, are emerging, integrated, and embedded in processes within organizations and related information system processes. From these processes, the immutability of the outcomes of decisions and actions, the development of formal and feasible models and methods for transparency, traceability, explainability, and the exploration of alternative approaches and

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In the field of scientific computing and models, the doctoral program deals with the construction of mathematical models using the tools of harmonic analysis, differential equations, number theory, and computer graphics, with the aim of solving problems in computer science, medicine, engineering, and other fields. In the area of numerical and symbolic methods, the doctoral program is structured around the theory and practice of numerical methods, approximation theory and optimization, and the mathematical background, methods, and applications of symbolic computation. Research in various related disciplines applying scientific computing, numerical or symbolic methods can also be included in the doctoral program. A wide range of topics such as signal and image processing, medical applications, engineering, shape recognition, computer vision, computer algebra applications, cryptography, and information security are being addressed.

Doctoral Program of Informatics Teaching Methodology

Head of the programme: Dr. Zoltán Illés, Habilitated Associate Professor

The aim of the doctoral program is to train research teachers with a broad knowledge of computer science, who are able to combine the discipline, its new findings, and pedagogical knowledge in a creative way, and who are able to perform demanding tasks in teaching, talent management, curriculum design and development, subject advising and subject management, as well as to provide a supply of new talent in the field of subject methodology in higher education. Research in educational methodologies needs to respond rapidly to a constantly changing knowledge base and to the impact of information technologies that are increasingly embedded in everyday life. The subject of informatics is changing very rapidly, with an increasing emphasis on programming alongside general informatics, as evidenced by the intensity of curricular debate at the international level. It is therefore essential for researchers to keep pace with the emergence of modern technologies and tools, not only at the user level but also with a broad, professional knowledge of programming. This complex task cannot be achieved without specific research skills.

Doctoral Program in Spatial Information and Spatial Data Science

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The doctoral program Spatial Informatics and Spatial Data Science is designed to analyze, interpret, and model spatial relationships of attribute data. It aims to develop a theoretical and practical understanding of spatial data technologies and their application, as well as research development to address the challenges in this field.

The program's thematic areas include remote sensing methods, web-GIS, navigation and mobile mapping technologies, spatial data visualization, spatial data mining, and other relevant topics in computer science.

Doctoral Program of IT Solutions in Engineering Sciences

Head of the programme: Dr. Levente Csóka, professor

Modern production technologies are inconceivable without IT research and development nowadays. The comprehensive, complex understanding and management of it is the driving force of modern societies.

The educational perspective within the framework of the doctoral program enables the acquisition of comprehensive proficiency at the interface points of the IT and engineering fields. Based on this knowledge, the students enable to conduct independent research, develop IT solutions and solve problems arising in engineering sciences.

The doctoral program provides theoretical and practical knowledge in the fields of informatics, which are essential for high-quality engineering solutions, including Industry 4.0 manufacturing technology, engineering materials science, mechanics and dynamic flow simulations.

IT research and development focuses on tasks whose solution supports the characteristics and multipurpose, applied use of metallic, polymeric and composite materials. Materials science issues range from molecular systems through an array of material properties to informatics and engineering design, finite element modeling and optimization procedures with IT systems. The background of these and similar engineering solutions is based on engineering mechanics.

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